



DOCUMENT VERSION	PRODUCTION HOST	SECURITY STATUS	PLATFORMS
v2.4.0 (Enterprise)	joypeoplehr.com	Active Hardened	Web + Flutter Mobile

1. Executive Summary

Joy PeopleHR (WorkForceOS Enterprise) is a comprehensive cloud Human Resource Management and Contractor Governance SaaS platform. It combines direct employee payroll, organizational hierarchy, statutory compliance, and third-party contractor manpower governance into a synchronized multi-platform architecture.

2. Architectural Comparison: HR Core vs. Vendor Governance

While both modules share similar data touchpoints (such as Attendance, Wages, and PF/ESI), their business personas and financial executions are distinct:

Functional Dimension	 HR Core (Internal Employees)	 Vendor Governance (Contract Staff)
Employment Status	Permanent, direct company payroll.	Employed by third-party staffing vendors.
Payroll Execution	Company calculates net pay and deposits directly to staff.	Vendor pays staff; Company pays Vendor's monthly invoice .
Attendance Purpose	Calculates monthly staff pay, paid leaves, and deductions.	Gate Pass & Audit: Verifies actual contractor presence to prevent billing inflation.
PF & ESI Compliance	Direct compliance filed directly to EPFO/ESIC by the company.	Audit Radar: Verifies vendor filed challans before approving final invoice payout (CLRA compliance).
Settlement Flow	Direct Payslips (Form 16).	3-Way Match: Purchase Order + Gate Hours + Vendor Invoice.

3. The 4–Stage Vendor Governance Lifecycle

STAGE 1: Onboarding & Licenses → Labor Licenses (OCR Scanner), Form V & Site Rosters STAGE 2: Operations & Time → Gate Attendance, Shift Logs & Overtime Audits STAGE 3: Settlement & 3-Way → Purchase Orders + Actual Wages + Vendor Invoice Match STAGE 4: Compliance & Audit → PF/ESI Challan Audits, Statutory Returns & Payout Approval

- 1.  **Vendor & Workforce Hub:** Vendor profiles, rate cards, OCR labor license tracker, and contractor site deployments.
- 2.  **Contractor Time & Attendance:** Biometric/RFID gate check-ins and overtime discrepancy audits.
- 3.  **Settlement & 3-Way Match Engine:** Automated validation matching (PO + Verified Attendance + Vendor Invoice).
- 4.  **Statutory & Compliance Radar:** Form V returns, PF/ESI challan verification, and audit trail logs.

4. Security Architecture & Anti–Session–Hijack Defense

To protect the SaaS production environment against session exfiltration, token theft, and cross–device copying:

- **Cryptographic Device Fingerprint Binding:** Generates a hybrid DJB2 + FNV–1a hardware signature from User–Agent, platform, screen resolution, timezone, and language. If tokens are moved to an unauthorized computer, the session is **instantly purged**.
- **Ephemeral SaaS Storage:** Avoids unmanaged persistent local storage in favor of tab–isolated sessionStorage and in–memory Supabase sessions.
- **Enterprise Security Headers:** Enforces X–Frame–Options: SAMEORIGIN, X–Content–Type–Options: nosniff, Strict–Transport–Security (HSTS), and strict Referrer Policy on server.js.
- **Dynamic CORS Origin Validation:** Strict whitelist restricting access exclusively to https://joypeoplehr.com and mobile application clients.

5. Multi–Platform Real–Time Sync (Web + Flutter)

Both the Web Enterprise App and the Flutter Mobile App connect to the same remote **Supabase Enterprise Cloud Database** (<https://wmqjmyzzamgxyeutbki.supabase.co>). Attendance clock–ins, leave approvals, and employee profile updates replicate in real–time across both platforms.

6. Live Production & CI/CD Status

Component	Configuration / Target	Status
Live Production URL	https://joypeoplehr.com	Online & Verified
Hosting Platform	Hostinger Cloud Node.js Runtime	Express Server Active
CI/CD Workflow	GitHub Actions (.github/workflows/ci.yml)	Automated Build & Typecheck
Health Check Endpoint	https://joypeoplehr.com/api/health	security_engine: active